

Computer Vision - 2026

Week #09. Face Recognition

Lectures by Alexei Kornaev ^{1,2}

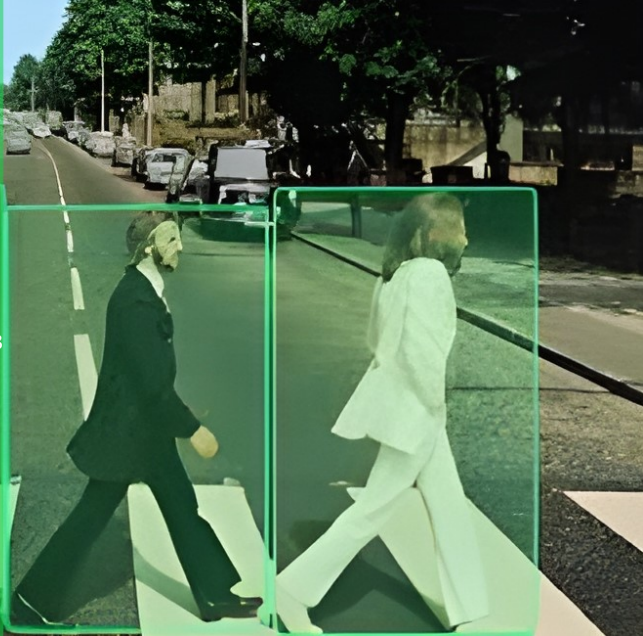
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Agenda

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- 1 Outcomes
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Section 1. Outcomes

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By the end of this week, students will be able to:

- 1 Distinguish between face verification (1:1 matching) and face identification (1:N search).
- 2 Explain the modern face recognition pipeline: **detection/alignment** → **embedding model** → **similarity scoring** → **thresholding/search**.
- 3 Compare major loss families for learning face embeddings: **contrastive loss**, **triplet loss**, and **angular-margin losses** such as ArcFace.
- 4 Understand why **image quality** matters and how modern methods such as quality-aware margins improve robustness.
- 5 Evaluate a face recognition system using **ROC**, **TAR/FAR**, **EER**, and **Rank-1 / CMC**, and discuss practical trade-offs.
- 6 Discuss limitations and risks related to pose, occlusion, domain shift, fairness, privacy, spoofing, and deepfakes.

Key takeaway: Modern face recognition is primarily an **embedding learning + similarity decision** problem, where data quality, evaluation protocol, and deployment constraints matter as much as raw accuracy.

Modern Face Recognition Pipeline

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Core idea

Modern face recognition systems do not usually predict an identity directly. Instead, they learn a compact embedding where images of the same person are close and different identities are far apart.

- ① **Face detection and alignment:** detect the face and normalize geometry.
- ② **Embedding model:** encode the aligned face into a feature vector.
- ③ **Normalization:** often use L_2 -normalized embeddings.
- ④ **Similarity scoring:** cosine similarity or Euclidean distance between embeddings.
- ⑤ **Decision:**
 - **Verification:** compare score with a threshold.
 - **Identification:** search in a gallery and return top- k matches.

Training vs inference:

- During training: margin-based metric/classification losses shape the embedding space.
- During inference: similarity scoring and threshold selection determine behavior.

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Section 2. Challenges in FR

Challenges in Face Recognition

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Overview of Challenges

Face recognition systems face several challenges that can significantly impact their performance. These include pose variation, illumination changes, occlusion, ethical concerns, and adversarial attacks.

- **Pose Variation:** Faces can appear in different orientations, making it difficult to match them.
- **Illumination Changes:** Variations in lighting conditions can alter the appearance of faces.
- **Occlusion:** Parts of the face may be obscured by objects or other faces.
- **Ethics:** Privacy concerns and biases in face recognition systems.
- **Adversarial Attacks:** Deliberate attempts to fool face recognition systems.

Pose Variation

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Pose Variation

Pose variation refers to the different orientations of a face relative to the camera. This can include frontal, profile, and oblique views.

- **Impact:** Reduces the accuracy of face recognition systems.
- **Solutions:** Use of 3D models, multi-view training data, and pose-invariant features.

Illumination Changes

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Illumination Changes

Illumination changes refer to variations in lighting conditions that can alter the appearance of a face.

- **Impact:** Can cause significant changes in the appearance of facial features.
- **Solutions:** Normalization techniques, use of infrared imaging, and deep learning models trained under various lighting conditions.

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Occlusion occurs when parts of a face are obscured by objects (e.g., masks, sunglasses), other people, or environmental factors. It degrades recognition accuracy by hiding critical facial features.

Understanding Occlusion



Figure: Examples of occluded face images from the MAFA dataset [Zeng et al., 2021].

- **Impact:**
 - Loss of key features (e.g., eyes, mouth).
 - State-of-the-art models like DeepFace or ArcFace suffer $>30\%$ accuracy drop under heavy occlusion.
- **Common Occlusions:** *Masks* (e.g., medical masks). *Accessories* (e.g., sunglasses, scarves). *Environmental* (e.g., hands, hair).
- **Challenges:**
 - Partial face data limits recognition methods.
 - Dynamic occlusions require real-time adaptation.

Solutions for Occlusion Handling

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Technical Approaches

Addressing occlusion requires robust feature extraction and reconstruction techniques.

- **Partial Face Recognition:**
 - Train models on cropped facial regions (e.g., eyes-only, mouth-only). Use *metric learning* to match partial and holistic embeddings.
- **Attention Mechanisms:**
 - Focus on visible regions, e.g. *Transformer Networks*) may prioritize non-occluded areas with attention [Liu et al., 2023].
- **Generative Models:**
 - Reconstruct occluded regions using GANs, Diffusion models.

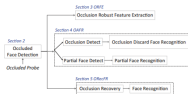


Figure: FR under occlusion [Zeng et al., 2021].

Responsible Deployment: Fairness, Privacy, Regulation

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Important distinction

A benchmark score does not automatically justify deployment in a real system.

- **Fairness:** error rates may differ across demographic groups, capture conditions, and thresholds.
- **Privacy:** face data are biometric data and require strict handling, storage, and consent policies.
- **Evaluation discipline:** claims about fairness or robustness should be based on protocol-specific audits, not only on one aggregate accuracy number.
- **Regulation:** biometric applications are increasingly governed by explicit legal constraints and risk categories.

Teaching point:

- what the model predicts,
- how the system is evaluated, and
- whether the deployment is ethically and legally acceptable.

Security Threats in Face Recognition

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Beyond ordinary test error

A face recognition system may fail not only because of natural variation, but also because of deliberate attacks.

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- **Presentation attacks / spoofing:** printed photos, replayed videos, masks.
- **Morphing attacks:** manipulated images designed to match multiple identities.
- **Adversarial perturbations:** inputs modified to change matching behavior.
- **Model inversion / privacy leakage:** attempts to recover sensitive information from model outputs or parameters.

Defenses

- liveness / presentation-attack detection,
- robust training and data hygiene,
- threshold selection under realistic attack models,
- privacy-aware system design.

Adversarial Attacks: Threat Models

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Adversarial Attacks in Face Recognition

Adversarial attacks manipulate inputs to deceive face recognition systems. Common types include:

- **Evasion Attacks:**
 - *Goal:* Fool the system during inference (e.g., adversarial glasses).
- **Poisoning Attacks:**
 - *Goal:* Corrupt training data (e.g. injecting malicious samples).
- **Model Inversion Attacks:**
 - *Goal:* Reconstruct training data from model outputs (privacy breach).

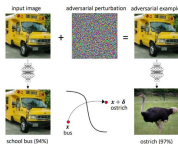


Figure: Adversarial noise attack.

Defending Against Adversarial Attacks

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Robustness Strategies

Mitigating adversarial attacks requires proactive defenses and robust model design.

- **Adversarial Training:**
 - Train models on adversarial examples to improve robustness.
 - *Limitation:* Computationally expensive; does not generalize to all attacks.
- **Input Preprocessing:**
 - Apply noise reduction, quantization, or JPEG compression to inputs.
- **Detection Mechanisms:**
 - Deploy detectors to flag adversarial inputs (e.g., using inconsistency in feature space).
- **Certified Defenses:**
 - Use mathematically proven robust models (e.g., randomized smoothing).

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Section 3. Traditional FR Methods

Traditional Face Recognition Methods

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Early approaches relied on dimensionality reduction and handcrafted features.

Linear Methods

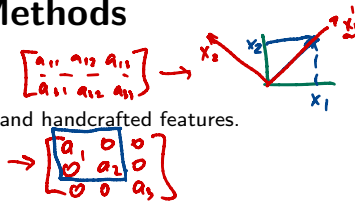


Figure: Eigenfaces [Turk and Pentland, 1991].

- **Eigenfaces [Turk and Pentland, 1991]:**
 - Uses PCA to project faces into a low-dimensional subspace.
 - Maximizes variance to capture dominant facial features.
- **LBPH (Local Binary Patterns Histogram):**
 - Encodes texture patterns using local binary operators.
 - Robust to monotonic illumination changes but struggles with pose/occlusion.

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Section 4. Deep Learning for FR

Deep Embedding Models for Face Recognition

Modern face recognition models learn discriminative face embeddings and compare them with similarity scores.

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Main families

- 6 Siamese / contrastive models [Koch et al., 2015]: learn similarity from positive and negative pairs.
- Triplet-based models (FaceNet) [Schroff et al., 2015]: enforce anchor-pos. pairs to be closer than anchor-neg. pairs by a margin.
- Angular-margin softmax models (ArcFace) [Deng et al., 2019]: optimize class separation directly on a normalized hypersphere.

Current practice:

- normalized embeddings;
- cosine similarity at inference;
- margin-based classification losses for scalable training.

Why ArcFace-style losses became dominant: they are usually easier to optimize at large scale than explicit pair/triplet mining while still producing strong embeddings.



Figure 2. **Model structure.** Our network consists of a batch input layer and a deep CNN followed by L_2 normalization, which results in the face embedding. This is followed by the triplet loss during training.



Figure 3. The **Triplet Loss** minimizes the distance between an anchor and a positive, both of which have the same identity, and maximizes the distance between the anchor and a negative of a different identity.

Figure: Face embedding intuition [Schroff et al., 2015; Deng et al., 2019].

Quality-Aware Face Embeddings

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Why quality matters

Low-quality faces (blur, low resolution, bad lighting, extreme pose, occlusion) are much harder to recognize reliably than clean frontal images.

- **MagFace [Meng et al., 2021]:** learns embeddings whose magnitude is related to face quality, helping the model organize easy and hard samples more meaningfully.
- **AdaFace [Kim et al., 2022]:** adapts the margin according to image quality, improving recognition on degraded and unconstrained imagery.
- **Practical lesson:** modern FR is not only about identity separation, but also about estimating when the input is reliable enough for matching.

Loss Functions in Face Recognition



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Metric Learning Objective

A loss enforces that embeddings of the same identity are closer than those of different identities.

- **Contrastive Loss :**

$$\mathcal{L} = \frac{1}{2N} \sum_{i=1}^N y_i \cdot d_i^2 + (1 - y_i) \cdot \max(\alpha - d_i, 0)^2$$

where $y_i = 1$ for same-class pairs, d_i =embedding distance, α =margin.

- **Triplet Loss [Schroff et al., 2015]:**

$$\mathcal{L} = \sum_{i=1}^N \max \left(\underbrace{\|f(a_i) - f(p_i)\|}^2 - \|f(a_i) - f(n_i)\|^2 + \underline{\alpha}, 0 \right)$$

- **ArcFace Loss [Deng et al., 2019]:**

$$L = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s \cos(\theta_{y_i} + m)}}{e^{s \cos(\theta_{y_i} + m)} + \sum_{j \neq y_i} e^{s \cos(\theta_j)}}$$

Hands-on Demo: Face Verification (1:1)

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Goal

Implement a minimal face recognition system using a pretrained embedding model and cosine similarity.

Task

- 1 Upload three face images:
 - two images of the **same person**
 - one image of a **different person**
- 2 Compute embeddings using a pretrained model (FaceNet).
- 3 Compute cosine similarity between embeddings.
- 4 Choose a threshold and implement a **verification decision**:

same person if $s > \tau$

Notebook

[Hands-on Face Embeddings \(Colab notebook\)](#)

Concept illustrated

- embedding representation
- cosine similarity



INNOPOLIS threshold-based verification (1:1)
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Evaluation Protocols and Metrics

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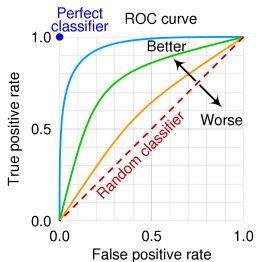


Figure: ROC-style evaluation for verification.

Two main tasks: Verification (1:1) and Identification (1:N)

Verification metrics

- **FAR** — false acceptance rate
- **FRR** — false rejection rate
- **TAR@FAR** — true accept rate at a fixed low FAR
- **EER** — equal error rate

Identification metrics

- **Rank-1 accuracy**
- **CMC curve**

Benchmarks

- **LFW**: important historical benchmark
- **IJB-B / IJB-C**: more realistic unconstrained evaluation
- **NIST FRVT**: large-scale operational testing

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Section 5. Conclusion & Discussion

Conclusion & Discussion

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Main technical lesson

- Face recognition is primarily an **embedding learning** problem solved with similarity-based decisions.
- Modern systems usually rely on **normalized embeddings + angular-margin training + thresholded matching**.

What matters in practice

- evaluation protocol: verification vs identification,
- operating point: security vs usability,
- image quality: blur, pose, occlusion, low resolution,
- deployment constraints: fairness, privacy, spoofing, regulation.

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